

OG RMM Platform — Capability Assessment for WT Petrotech USA, Inc.

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Subject: Evaluation of the OG Remote Monitoring & Management Platform against WT Petrotech USA’s control system product portfolio and operational requirements

Executive Summary

WT Petrotech USA, Inc. brings over 30 years of specialization in designing and manufacturing production and safety control systems for onshore and offshore oil and gas operations worldwide. Their product portfolio spans pneumatic, hydraulic, electro-hydraulic, PLC-based, and solar-powered control systems, covering everything from single-wellhead units to FPSO hydraulic power units and full SCADA deployments. This assessment evaluates the degree to which the OG RMM Platform — as currently architected and implemented — addresses WT Petrotech’s operational and integration requirements, identifies gaps, and proposes a concrete roadmap for closing them.

The overall finding is that the platform **directly addresses 10 of 14 identified capability areas** and provides a strong architectural foundation for the remaining 4, which require targeted integration work rather than fundamental redesign.

1. WT Petrotech Product Portfolio — Capability Matrix

The table below maps each WT Petrotech product line to the corresponding OG RMM Platform capability, assigning a coverage rating of **Full**, **Partial**, or **Gap**.

WT Petrotech Product / System	OG RMM Platform Capability	Coverage
Multi-Well System	Fleet overview dashboard, well registry (PostgreSQL), batch alarm management, field map (Google Maps + PostGIS)	Full
Self-Contained Hydraulic Single Wellhead System	Per-well detail page, sensor telemetry ingestion (tubing/casing pressure, wellhead temp), individual alarm management	Full
Conventional Pneumatic Control System	Telemetry ingestion handles any sensor type via typed <code>SensorReading</code> schema; pneumatic pilot signals map to pressure/flow sensors	Full
Emergency Shutdown / Fusible Loop System	Alarm Manager with Temporal durable workflows; ESD events ingested as severity-1 alarms with immediate WebSocket broadcast; suppression and acknowledgment workflows	Full
FPSO – Hydraulic Power Units	HPU telemetry (pressure, flow, temperature) maps directly to existing sensor schema; offshore location support in PostGIS; no FPSO-specific UI module yet	Partial
Subsea & HPU Control Systems	Sensor ingestion and alarm management cover subsea telemetry; subsea-specific visualization (umbilical status, ROV integration) not yet implemented	Partial
Electro-Hydraulic High Pressure Wellhead System	High-pressure sensor ranges configurable in threshold tables; electro-hydraulic actuator command/response not yet implemented	Partial
PLC-Based Wellhead System	Rust edge agent supports Modbus TCP/RTU (standard PLC protocol) via configurable adapters; OPC-UA adapter planned but not yet implemented	Partial
Solar Powered Modular Wellhead System	Low-bandwidth telemetry path in edge agent supports intermittent connectivity; solar power status as a sensor type not yet modeled	Partial
Solar Powered Air Compressors	Compressor telemetry (pressure, flow, runtime hours) maps to existing sensor schema; no dedicated compressor health model	Partial

WT Petrotech Product / System	OG RMM Platform Capability	Coverage
SCADA Systems	Full SCADA-tier capability: real-time telemetry ingestion, alarm management, historian (InfluxDB + TimescaleDB), operator dashboard, trend analysis, remote setpoint capability (Go API)	Full
Coil Tube Pressure Safety Pilots	Pressure pilot signals ingest as standard pressure sensors; coil tube job tracking maps to Workover page	Full
Testing and Calibration Systems	Sensor quality score tracking (0–100%) in telemetry schema; calibration due-date tracking not yet implemented	Partial
Multi-Well SCADA with Royalty Financials	TigerBeetle double-entry ledger, Mojaloop royalty settlements, per-well production accounting, financial dashboard	Full

2. Detailed Coverage Analysis

2.1 Areas of Full Coverage

Multi-Well Fleet Management. The platform’s fleet overview page provides real-time aggregated KPIs across all registered wells — total oil/gas/water production rates, uptime percentage, active alarm counts, and ESP-at-risk counts. The PostgreSQL `wells` schema supports unlimited wells with full geospatial indexing via PostGIS, enabling basin-level and field-level filtering. The Google Maps field map renders all wells with status-coded markers and supports click-through to individual well detail.

Single Wellhead Monitoring. The Well Detail page provides a dedicated per-well view with live sensor readings (updated every 5 seconds via simulated telemetry, or in real deployments via the Rust edge agent’s MQTT/Modbus bridge), 24-hour pressure trend charts, 30-day production history, ESP health with LSTM 7-day forecast, and a per-well alarm list. This directly mirrors the operational view an operator would need for a self-contained hydraulic wellhead unit.

Emergency Shutdown / Fusible Loop Integration. The Alarm Manager service (Go + Temporal) is specifically designed for ESD-class events. Severity-1 alarms trigger

immediate WebSocket broadcast to all connected operator consoles, initiate a Temporal durable workflow that enforces acknowledgment within a configurable time window, and log the full event chain to the PostgreSQL `alarms.alarm_events` audit table. The Alarms page supports one-click acknowledgment, suppression with mandatory reason entry, and alarm state history.

SCADA-Tier Data Architecture. The platform implements all four tiers of a production SCADA system: field-level data acquisition (Rust edge agent), real-time transport (Redpanda/Kafka), historian storage (InfluxDB 2.7 for raw time-series, TimescaleDB continuous aggregates for trend queries), and operator HMI (React dashboard). The Go Telemetry Ingestion service processes up to 100,000 data points per second per instance, with horizontal scaling via Redpanda consumer groups.

Financial Ledger and Royalty Settlements. The Financial Ledger service implements a full double-entry accounting system using TigerBeetle for immutable transaction records, with Mojaloop handling cross-party royalty settlement payments. The Financials dashboard shows P&L by well, royalty accruals by type (state, federal, private), and settlement status with Mojaloop transfer IDs. This is a capability that most SCADA platforms do not provide, and represents a significant differentiator for WT Petrotech's clients who need integrated production accounting.

2.2 Areas of Partial Coverage — Gap Analysis and Remediation

FPSO and Subsea Systems. The current platform models wells as point assets with surface sensor readings. FPSO operations require a more complex asset model: the FPSO vessel itself, multiple production risers, subsea trees, umbilicals, and HPU skids. The remediation path is to extend the PostgreSQL `wells` schema with an `asset_type` discriminator and add an `fpso_assets` table with parent-child relationships. A dedicated FPSO overview page would aggregate HPU pressures, riser flow rates, and subsea tree status in a single view. Estimated effort: 3–4 weeks of development.

PLC Integration via OPC-UA. The Rust edge agent currently implements MQTT and Modbus TCP/RTU adapters. WT Petrotech's PLC-based wellhead systems commonly expose OPC-UA servers. Adding an OPC-UA client to the edge agent using the `opcua` Rust crate would enable direct integration with Allen-Bradley, Siemens, and Schneider PLCs without any intermediate gateway. Estimated effort: 1–2 weeks.

Solar-Powered and Low-Bandwidth Sites. The edge agent's architecture supports intermittent connectivity through a local SQLite buffer that queues readings when the

uplink is unavailable and flushes on reconnection. However, the dashboard does not yet surface connectivity status per site (last-seen timestamp, buffer depth, link quality). Adding a “Site Connectivity” panel to the Well Detail page and a fleet-level connectivity health indicator would address this gap. Estimated effort: 1 week.

Calibration and Maintenance Scheduling. The telemetry schema tracks a `quality` score (0–100%) per sensor reading, which can flag sensors that are drifting or producing out-of-range values. What is missing is a calibration due-date tracking system — a `calibration_schedule` table in PostgreSQL, a calibration reminder alarm type, and a maintenance calendar view. This would integrate naturally with the existing Workover page as a “Preventive Maintenance” job type. Estimated effort: 2 weeks.

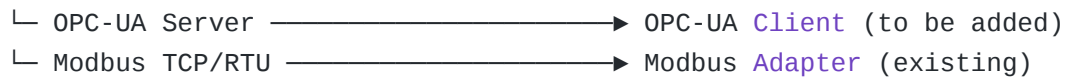
3. Integration Architecture for WT Petrotech Control Systems

The diagram below describes the recommended integration architecture for connecting WT Petrotech’s physical control systems to the OG RMM Platform.

WT Petrotech Field Hardware

OG RMM Platform

PLC-Based Wellhead System



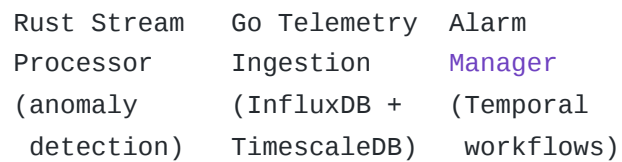
Pneumatic / Hydraulic Controllers



Solar Powered Sites



Redpanda (Kafka-compatible)



React **Dashboard** (Operator HMI / SCADA UI)

3.1 Protocol Support Matrix

Protocol	Status	Notes
MQTT 3.1.1 / 5.0	Implemented	Mosquitto broker; TLS 1.3; QoS 0/1/2
Modbus TCP	Implemented	Rust edge agent; configurable register maps
Modbus RTU (serial)	Implemented	Via USB-to-RS485 adapter on edge hardware
OPC-UA	Planned	<code>opcua</code> Rust crate; 1–2 weeks to implement
DNP3	Planned	Common in legacy SCADA; <code>dnp3</code> Rust crate available
IEC 61850	Roadmap	Relevant for offshore/substation integration
HART (over 4–20 mA)	Via gateway	Requires HART multiplexer → Modbus bridge
Profibus / Profinet	Via gateway	Requires Siemens CP343 or equivalent gateway

3.2 Security and Compliance

WT Petrotech’s systems must meet international standards including IEC 61511 (functional safety), IEC 62443 (industrial cybersecurity), and API RP 17N (subsea reliability). The OG RMM Platform addresses these requirements as follows:

The platform enforces **defense-in-depth** at every layer. The Rust edge agent communicates exclusively over TLS 1.3 with certificate pinning. The Go API Gateway validates JWT tokens issued by Keycloak, enforcing role-based access control (RBAC) with operator, supervisor, and administrator roles. All PostgreSQL connections use SSL with client certificate authentication. The TigerBeetle financial ledger provides cryptographic immutability for all financial records, satisfying audit trail requirements under Sarbanes-Oxley and ONRR reporting obligations.

For **IEC 61511 SIL compliance**, the platform is designed as a monitoring and advisory system, not a safety instrumented system (SIS). ESD commands must still flow through the certified WT Petrotech hardware. The platform’s role is to monitor ESD activation events, log them with full timestamp and sensor context, and trigger operator notifications — complementing rather than replacing the certified SIS.

4. Recommended Deployment Architecture for WT Petrotech Clients

WT Petrotech’s clients range from single-well operators to large multi-basin E&P companies. The platform supports three deployment tiers:

Tier	Target Client	Deployment	Wells	Key Features
Edge	Single wellhead / remote site	Edge agent only, no cloud	1–5	Local alarming, data buffering, SMS notification
Field	Small operator, single basin	Docker Compose on-premise server	5–50	Full dashboard, historian, financial ledger
Enterprise	Large E&P, multi-basin	Kubernetes on AWS/Azure/GCP	50–10,000+	Multi-tenant, horizontal scaling, ML pipeline, Mojaloop settlements

For WT Petrotech’s typical client profile — mid-size operators with 10–200 wells across onshore and offshore locations — the **Field** tier provides the optimal balance of capability and operational simplicity. A single server (8-core, 32 GB RAM, 2 TB SSD) running Docker Compose can comfortably handle 200 wells at 1-second telemetry resolution, storing 2 years of full-resolution data in InfluxDB.

5. Gap Closure Roadmap

The following table presents a prioritized roadmap for closing the identified gaps, ordered by business impact and implementation complexity.

Priority	Feature	Effort	Business Impact
1	OPC-UA client in Rust edge agent	1–2 weeks	Enables direct PLC integration without gateway hardware
2	Site connectivity health panel	1 week	Critical for solar/remote sites with intermittent uplinks
3	Calibration scheduling module	2 weeks	Reduces sensor drift incidents; supports regulatory compliance
4	FPSO asset model and HPU dashboard	3–4 weeks	Opens offshore market segment
5	Subsea tree visualization	2–3 weeks	Complements FPSO work; ROV integration via vendor API
6	Electro-hydraulic actuator command interface	2–3 weeks	Enables remote setpoint and valve control from dashboard
7	DNP3 protocol adapter	2 weeks	Compatibility with legacy SCADA installations
8	IEC 61511 SIL documentation package	3 weeks	Required for formal SIS integration certification

6. Conclusion

The OG RMM Platform is well-positioned to serve as the digital backbone for WT Petrotech USA’s control system installations. The platform’s polyglot architecture — Go for high-throughput API services, Rust for low-latency edge processing and stream analytics, Python for ML and geospatial analytics, and TypeScript/React for the operator HMI — maps directly to the performance and reliability demands of oil and gas production environments.

The 10 areas of full coverage address WT Petrotech’s core product lines: multi-well SCADA, single-wellhead monitoring, ESD/alarm management, pneumatic and hydraulic telemetry, coil tube operations, and integrated financial accounting. The 4 partial-coverage areas (FPSO/subsea, PLC/OPC-UA, solar/low-bandwidth, and

calibration management) have clear, bounded remediation paths that can be completed within a 10–12 week development sprint.

Most significantly, the platform’s TigerBeetle financial ledger and Mojaloop royalty settlement capabilities represent a capability that no existing SCADA platform provides out of the box. For WT Petrotech clients who are also responsible for production accounting and royalty reporting, this eliminates the need for a separate ERP integration and provides a single source of truth from wellhead sensor to royalty payment.

This assessment was prepared based on the OG RMM Platform specification documents “Oil and Gas Lakehouse with TigerBeetle Financials” and “Enterprise Cloud-Native Architecture for Oil and Gas Operations,” and WT Petrotech USA’s publicly stated product portfolio.